KERALA DIPLOMA CURRICULUM REVISION 2026

Course 3052 | 3 Credits: 3 | Program Core | Source-grounded

Manufacturing Technology for

 To provide a basic knowledge on metal casting and fabrication processes.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 3052 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 3052.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Automobile Engineering
Course code	3052
Course title	Manufacturing Technology for
Semester	3 Credits: 3
Course category	Program Core
Periods per week	3 (L:2, T:1, P:0) Periods per semester: 45
Periods per semester	45

Item	Official information
Credits	3
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

40 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To provide a basic knowledge on metal casting and fabrication processes.
- To acquire knowledge of machine tools for specific applications, metal forming,
- additive manufacturing and their applications.

Course outcomes

CO	Official outcome	Study level
CO1	Choose metal casting and powder metallurgy. 12 Applying Classify sheet metal works used in automobile	See official syllabus
CO2	manufacturing and relate a fabrication process for 11 Understanding specific automobile application. Demonstrate the working of standard machine tools such as lathe, shaping, milling, drilling	See official syllabus
CO3	11 Understanding machines and the basics of flexible manufacturing system. Outline the application of forging, rolling and	See official syllabus
CO4	additive manufacturing techniques in automobile 9 Understanding industry.	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 3
CO3	CO3 2
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Choose metal casting and powder metallurgy.	12	As specified
Module 2	particular fabrication process for specific automobile application.	11	As specified
Module 3	milling, drilling machine and the basics of flexible manufacturing system.	8	As specified
Module 4	:	9	As specified

MODULE 1

Choose metal casting and powder metallurgy.

This module is presented from the official syllabus scope for Manufacturing Technology for. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Choose metal casting and powder metallurgy.
- Official duration: As specified

- Official topic entries captured: 12
- The full source excerpt is preserved below for coverage checking.

Recall casting and pattern. 1 Remembering Outline the requirements of a pattern material

Recall casting and pattern. 1 Remembering Outline the requirements of a pattern material is an official topic in Module 1 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Recall casting and pattern. 1 Remembering Outline the requirements of a pattern material using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, recall casting and pattern. 1 remembering outline the requirements of a pattern material should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Recall casting and pattern. 1 Remembering Outline the requirements of a pattern material means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Recall casting and pattern. 1 Remembering Outline the requirements of a pattern material □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

1 Understanding and list different types of pattern materials. Classify patterns like single, split, loose piece

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Study points

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Engineering or professional relevance

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Study points

- Define or identify allowances such as shrinkage, machining, 1 Understanding draft, distortion, shake allowances. Define mould, cope, drag, runner, riser, core using standard technical terminology.
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Study points

- Define or identify 1 Understanding and chaplets. Distinguish green sand moulding and dry sand using standard technical terminology.
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Engineering or professional relevance

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Topic study guide

Aspect	What to prepare
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Engineering or professional relevance

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1 Understanding merits and demerits . Illustrate the methods for making of metal

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Study points

- Define or identify 1 Understanding merits and demerits . Illustrate the methods for making of metal using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
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Engineering or professional relevance

In Polytechnic practice, 1 understanding merits and demerits . illustrate the methods for making of metal should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

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> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] 1 Understanding operations in powder metallurgy process. Identify automobile componentsmade by Applying [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

1 powder metallurgy. Materials For Automobiles And Metal Casting Introduction to metal castings and patterns, pattern materials, types of patterns-single, split, loose piece, sweep, pattern allowances such as shrinkage, machining, draft, distortion, shake allowances. Moulding terminology such as cope, drag, runner, riser, core and chaplets. Definition green sand and dry sand moulding, permanent mould casting such as hot chamber and cold chamber die casting and centrifugal casting and identify automotive components made these processes. Discuss various casting defects. Powder metallurgy steps, secondary operations. Name the automotive components made by powder metallurgy, merits and demerits of powder metallurgy. Classify sheet metal works used in Automobile field and relate a

1 powder metallurgy. Materials For Automobiles And Metal Casting Introduction to metal castings and patterns, pattern materials, types of patterns-single, split, loose piece, sweep, pattern allowances such as shrinkage, machining, draft, distortion, shake allowances. Moulding terminology such as cope, drag, runner, riser, core and chaplets. Definition green sand and dry sand moulding, permanent mould casting such as hot chamber and cold chamber die casting and centrifugal casting and identify automotive components made these processes. Discuss various casting defects. Powder metallurgy steps, secondary operations. Name the automotive components made by powder metallurgy, merits and demerits of powder metallurgy. Classify sheet metal works used in Automobile field and relate a is an official topic in Module 1 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 powder metallurgy. Materials For Automobiles And Metal Casting Introduction to metal castings and patterns, pattern materials, types of patterns-single, split, loose piece, sweep, pattern allowances such as shrinkage, machining, draft, distortion, shake allowances. Moulding terminology such as cope, drag, runner, riser, core and chaplets. Definition green sand and dry sand moulding, permanent mould casting such as hot chamber and cold chamber die casting and centrifugal casting and identify automotive components made these

processes. Discuss various casting defects. Powder metallurgy steps, secondary operations. Name the automotive components made by powder metallurgy, merits and demerits of powder metallurgy. Classify sheet metal works used in Automobile field and relate a using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 powder metallurgy. materials for automobiles and metal casting introduction to metal castings and patterns, pattern materials, types of patterns-single, split, loose piece, sweep, pattern allowances such as shrinkage, machining, draft, distortion, shake allowances. moulding terminology such as cope, drag, runner, riser, core and chaplets. definition green sand and dry sand moulding, permanent mould casting such as hot chamber and cold chamber die casting and centrifugal casting and identify automotive components made these processes. discuss various casting defects. powder metallurgy steps, secondary operations. name the automotive components made by powder metallurgy, merits and demerits of powder metallurgy. classify sheet metal works used in automobile field and relate a should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 powder metallurgy. Materials For Automobiles And Metal Casting Introduction to metal castings and patterns, pattern materials, types of patterns-single, split, loose piece, sweep, pattern allowances such as shrinkage, machining, draft, distortion, shake allowances. Moulding terminology such as cope, drag, runner, riser, core and chaplets. Definition green sand and dry sand moulding, permanent mould casting such as hot chamber and cold chamber die casting and centrifugal casting and identify automotive components made these processes. Discuss various casting defects. Powder metallurgy steps, secondary operations. Name the automotive components made by powder metallurgy, merits and demerits of powder metallurgy. Classify sheet metal works used in Automobile field and relate a means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Choose metal casting and powder metallurgy.

  M1.01      Recall casting and pattern.                            1
Remembering


             Outline the requirements of a pattern material
  M1.02                                                             1
Understanding
             and list different types of pattern materials.
             Classify patterns like single, split, loose piece
  M1.03                                                             1
Understanding
             and sweep pattern.
             Define pattern allowance and define pattern
  M1.04      allowances such as shrinkage, machining,               1
Understanding
             draft, distortion, shake allowances.
             Define mould, cope, drag, runner, riser, core
  M1.05                                                             1
Understanding
             and chaplets.
             Distinguish green sand moulding and dry sand
  M1.06      moulding and identify automotive components            1
Applying
             made by these processes.
             Explain hot chamber die casting, cold
             chamber die casting, centrifugal casting and
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

particular fabrication process for specific automobile application.

This module is presented from the official syllabus scope for Manufacturing Technology for. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: particular fabrication process for specific automobile application.
- Official duration: As specified
- Official topic entries captured: 11
- The full source excerpt is preserved below for coverage checking.

1 Understanding application of sheet metal in automobile. List metals used for sheet metal work and

1 Understanding application of sheet metal in automobile. List metals used for sheet metal work and is an official topic in Module 2 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding application of sheet metal in automobile. List metals used for sheet metal work and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding application of sheet metal in automobile. list metals used for sheet metal work and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding application of sheet metal in automobile. List metals used for sheet metal work and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 1 Understanding application of sheet metal in automobile. List metals used for sheet metal work and [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

1 Understanding standard gauge numbers used for sheet metals. Define various sheet metal operations like - shearing, cutting off, parting, blanking,

1 Understanding standard gauge numbers used for sheet metals. Define various sheet metal operations like - shearing, cutting off, parting, blanking, is an official topic in Module 2 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

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Study points

- Define or identify 1 Understanding standard gauge numbers used for sheet metals. Define various sheet metal operations like - shearing, cutting off, parting, blanking, using standard technical terminology.
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Definition/identity	What 1 Understanding standard gauge numbers used for sheet metals. Define various sheet metal operations like - shearing, cutting off, parting, blanking, means in the official course context
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Malayalam support: Module 2-□□ 1 Understanding standard gauge numbers used for sheet metals. Define various sheet metal operations like - shearing, cutting off, parting, blanking, □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

1 Understanding punching, notching, slitting, lancing, bending, drawing and squeezing.

1 Understanding punching, notching, slitting, lancing, bending, drawing and squeezing. is an official topic in Module 2 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

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Study points

- Define or identify 1 Understanding punching, notching, slitting, lancing, bending, drawing and squeezing. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
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Engineering or professional relevance

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Topic study guide

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Definition/identity	What 1 Understanding punching, notching, slitting, lancing, bending, drawing and squeezing. means in the official course context
Study lens	Principle → components or stages → result → application
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Define welding and weldability. 1 Remembering Explain the features of welding and general

Define welding and weldability. 1 Remembering Explain the features of welding and general is an official topic in Module 2 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

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Study points

- Define or identify Define welding and weldability. 1 Remembering Explain the features of welding and general using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
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Engineering or professional relevance

In Polytechnic practice, define welding and weldability. 1 remembering explain the features of welding and general should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Define welding and weldability. 1 Remembering Explain the features of welding and general means in the official course context
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1 Understanding classification of welding. List the equipment used in arc welding and

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Study points

- Define or identify 1 Understanding classification of welding. List the equipment used in arc welding and using standard technical terminology.
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- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
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Engineering or professional relevance

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Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding classification of welding. List the equipment used in arc welding and means in the official course context
Study lens	Principle → components or stages → result → application
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1 Understanding discuss the arc welding principle. Outline MIG Welding, TIG Welding and

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Study points

- Define or identify 1 Understanding discuss the arc welding principle. Outline MIG Welding, TIG Welding and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
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Engineering or professional relevance

In Polytechnic practice, 1 understanding discuss the arc welding principle. outline mig welding, tig welding and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding discuss the arc welding principle. Outline MIG Welding, TIG Welding and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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1 Understanding Submerged arc welding. List equipment for gas welding and gas

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Concept and explanation

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Study points

- Define or identify 1 Understanding Submerged arc welding. List equipment for gas welding and gas using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding submerged arc welding. list equipment for gas welding and gas should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding Submerged arc welding. List equipment for gas welding and gas means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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1 Understanding cutting. Illustrate oxy-acetylene welding and types of

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Study points

- Define or identify 1 Understanding cutting. Illustrate oxy-acetylene welding and types of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
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Engineering or professional relevance

In Polytechnic practice, 1 understanding cutting. illustrate oxy-acetylene welding and types of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding cutting. Illustrate oxy-acetylene welding and types of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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1 Understanding flames. Explain resistance welding such as – spot,

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Concept and explanation

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Study points

- Define or identify 1 Understanding flames. Explain resistance welding such as – spot, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding flames. explain resistance welding such as – spot, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding flames. Explain resistance welding such as – spot, means in the official course context
Study lens	Principle → components or stages → result → application
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Malayalam support: Module 2-[illegible] 1 Understanding flames. Explain resistance welding such as – spot, [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

1 Understanding butt, seam welding.

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Concept and explanation

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Study points

- Define or identify 1 Understanding butt, seam welding. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
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Engineering or professional relevance

In Polytechnic practice, 1 understanding butt, seam welding. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding butt, seam welding. means in the official course context
Study lens	Principle → components or stages → result → application
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Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 1 Understanding butt, seam welding. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Comparesoldering and brazing. 1 Understanding Sheet Metal And Fabrication Processes Introduction to Sheet Metal Work and Applications, metals used for Sheet Metal work, Standard Gauge numbers. Sheet metal operations-shearing, cutting off, Parting, blanking, Punching, notching, slitting, Lancing, Bending, Drawing and squeezing. Introduction to Welding - Classification of welding such as fusion welding and solid state welding. - Arc welding principle and equipment, Types of Arc welding - MIG Welding, TIG Welding, Submerged arc welding. Equipment for Gas welding and gas cutting. Oxy - acetylene welding and Types of flames. Principle of resistance welding - Spot, butt, seam Welding. Comparison of soldering and Brazing. Demonstrate the working of standard machine tools like lathe, shaping,

Comparesoldering and brazing. 1 Understanding Sheet Metal And Fabrication Processes Introduction to Sheet Metal Work and Applications, metals used for Sheet Metal work, Standard Gauge numbers. Sheet metal operations-shearing, cutting off, Parting, blanking, Punching, notching, slitting, Lancing, Bending, Drawing and squeezing. Introduction to Welding - Classification of welding such as fusion welding and solid state welding. - Arc welding principle and equipment, Types of Arc welding - MIG Welding, TIG Welding, Submerged arc welding. Equipment for Gas welding and gas cutting. Oxy - acetylene welding and Types of flames. Principle of resistance welding - Spot, butt, seam Welding. Comparison of soldering and Brazing. Demonstrate the working of standard machine tools like lathe, shaping, is an official topic in Module 2 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Comparesoldering and brazing. 1 Understanding Sheet Metal And Fabrication Processes Introduction to Sheet Metal Work and Applications, metals used for Sheet Metal work, Standard Gauge numbers. Sheet metal operations-shearing, cutting off, Parting, blanking, Punching, notching, slitting, Lancing, Bending, Drawing and squeezing. Introduction to Welding - Classification of welding such as fusion welding and solid state welding. - Arc welding principle and equipment, Types of Arc welding - MIG Welding, TIG Welding, Submerged arc welding. Equipment for Gas welding and gas cutting. Oxy - acetylene welding and

Types of flames. Principle of resistance welding - Spot, butt, seam Welding. Comparison of soldering and Brazing. Demonstrate the working of standard machine tools like lathe, shaping, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, comparesoldering and brazing. 1 understanding sheet metal and fabrication processes introduction to sheet metal work and applications, metals used for sheet metal work, standard gauge numbers. sheet metal operations-shearing, cutting off, parting, blanking, punching, notching, slitting, lancing, bending, drawing and squeezing. introduction to welding - classification of welding such as fusion welding and solid state welding. - arc welding principle and equipment, types of arc welding - mig welding, tig welding, submerged arc welding. equipment for gas welding and gas cutting. oxy - acetylene welding and types of flames. principle of resistance welding - spot, butt, seam welding. comparison of soldering and brazing. demonstrate the working of standard machine tools like lathe, shaping, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Comparesoldering and brazing. 1 Understanding Sheet Metal And Fabrication Processes Introduction to Sheet Metal Work and Applications, metals used for Sheet Metal work, Standard Gauge numbers. Sheet metal operations-shearing, cutting off, Parting, blanking, Punching, notching, slitting, Lancing, Bending, Drawing and squeezing. Introduction to Welding - Classification of welding such as fusion welding and solid state welding. - Arc welding principle and equipment, Types of Arc welding - MIG Welding, TIG Welding, Submerged arc welding. Equipment for Gas welding and gas cutting. Oxy - acetylene welding and Types of flames. Principle of resistance welding - Spot, butt, seam Welding. Comparison of soldering and Brazing. Demonstrate the working of standard machine tools like lathe, shaping, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Comparesoldering and brazing. 1 Understanding Sheet Metal And Fabrication Processes Introduction to Sheet Metal Work and Applications, metals used for Sheet Metal work, Standard Gauge numbers. Sheet metal operations-shearing, cutting off, Parting, blanking, Punching, notching, slitting, Lancing, Bending, Drawing and squeezing. Introduction to Welding - Classification of welding such as fusion welding and solid state welding. - Arc welding principle and equipment, Types of Arc welding - MIG Welding, TIG Welding, Submerged arc welding. Equipment for Gas welding and gas cutting. Oxy - acetylene welding and Types of flames. Principle of resistance welding - Spot, butt, seam Welding. Comparison of soldering and Brazing. Demonstrate the working of standard machine tools like lathe, shaping, □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
particular fabrication process for specific automobile application.

             Define sheet metal and identify the areas of
  M2.01                                                         1
Understanding
             application of sheet metal in automobile.

             List metals used for sheet metal work and
  M2.02                                                         1
Understanding
             standard gauge numbers used for sheet metals.
             Define various sheet metal operations like -
             shearing, cutting off, parting, blanking,
  M2.03                                                         1
Understanding
             punching, notching, slitting, lancing, bending,
             drawing and squeezing.

  M2.04      Define welding and weldability.                    1
Remembering

             Explain the features of welding and general
  M2.05                                                         1
Understanding
             classification of welding.

             List the equipment used in arc welding and
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

milling, drilling machine and the basics of flexible manufacturing system.

This module is presented from the official syllabus scope for Manufacturing Technology for. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: milling, drilling machine and the basics of flexible manufacturing system.
- Official duration: As specified
- Official topic entries captured: 8
- The full source excerpt is preserved below for coverage checking.

their functions, working and lathe 2 Understanding specifications. Define various lathe operations such as facing, centering, drilling, plain turning, boring,

their functions, working and lathe 2 Understanding specifications. Define various lathe operations such as facing, centering, drilling, plain turning, boring, is an official topic in Module 3 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify their functions, working and lathe 2 Understanding specifications. Define various lathe operations such as facing, centering, drilling, plain turning, boring, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, their functions, working and lathe 2 understanding specifications. define various lathe operations such as facing, centering, drilling, plain turning, boring, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What their functions, working and lathe 2 Understanding specifications. Define various lathe operations such as facing, centering, drilling, plain turning, boring, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] their functions, working and lathe 2 Understanding specifications. Define various lathe operations such as facing, centering, drilling, plain turning, boring, [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

reaming, chamfering taper turning and thread cutting. Identify the components of a of shaper and demonstrate the crank and slotted lever type 1 Understanding M3.03 quick return motion mechanism.

reaming, chamfering taper turning and thread cutting. Identify the components of a of shaper and demonstrate the crank and slotted lever type 1 Understanding M3.03 quick return motion mechanism. is an official topic in Module 3 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify reaming, chamfering taper turning and thread cutting. Identify the components of a of shaper and demonstrate the crank and slotted lever type 1 Understanding M3.03 quick return motion mechanism. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, reaming, chamfering taper turning and thread cutting. identify the components of a of shaper and demonstrate the crank and slotted lever type 1 understanding m3.03 quick return motion mechanism. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What reaming, chamfering taper turning and thread cutting. Identify the components of a of shaper and demonstrate the crank and slotted lever type 1 Understanding M3.03 quick return motion mechanism. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ reaming, chamfering taper turning and thread cutting. Identify the components of a of shaper and demonstrate the crank and slotted lever type 1 Understanding M3.03 quick return motion mechanism. □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

Identify the components of a drilling machine.

Identify the components of a drilling machine. is an official topic in Module 3 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify the components of a drilling machine. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify the components of a drilling machine. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Identify the components of a drilling machine. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Identify the components of a drilling machine. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Define drilling, reaming and boring. 1 Understanding Identify the components of a Column and

Define drilling, reaming and boring. 1 Understanding Identify the components of a Column and is an official topic in Module 3 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Define drilling, reaming and boring. 1 Understanding Identify the components of a Column and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, define drilling, reaming and boring. 1 understanding identify the components of a column and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Define drilling, reaming and boring. 1 Understanding Identify the components of a Column and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Malayalam support: Module 3-□□ Define drilling, reaming and boring. 1 Understanding Identify the components of a Column and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

2 Understanding knee type milling machine. Define plain milling, side milling, face

2 Understanding knee type milling machine. Define plain milling, side milling, face is an official topic in Module 3 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding knee type milling machine. Define plain milling, side milling, face using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding knee type milling machine. define plain milling, side milling, face should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding knee type milling machine. Define plain milling, side milling, face means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 2 Understanding knee type milling machine. Define plain milling, side milling, face [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

1 Understanding milling, angle milling, end milling.

1 Understanding milling, angle milling, end milling. is an official topic in Module 3 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding milling, angle milling, end milling. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding milling, angle milling, end milling. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding milling, angle milling, end milling. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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Malayalam support: Module 3-[illegible] 1 Understanding milling, angle milling, end milling. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Identify the components of a CNC machine. 1 Understanding Outline the Flexible manufacturing system

Identify the components of a CNC machine. 1 Understanding Outline the Flexible manufacturing system is an official topic in Module 3 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify the components of a CNC machine. 1 Understanding Outline the Flexible manufacturing system using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify the components of a cnc machine. 1 understanding outline the flexible manufacturing system should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Identify the components of a CNC machine. 1 Understanding Outline the Flexible manufacturing system means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Identify the components of a CNC machine. 1 Understanding Outline the Flexible manufacturing system □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

1 Understanding and relate its applications. Machine Tools And Operations Lathe - constructional features and working, various lathe operations such as facing, centering, drilling, plain turning, boring, reaming, chamfering taper turning and thread cutting. - Shaper - constructional features and working of the crank and slotted lever type quick return motion mechanism - constructional features of a drilling machine. Hole making operations - drilling - reaming - boring. Milling - Column and knee type milling machine - various operations performed on milling machine. Constructional features of CNC machines, illustrate flexible manufacturing system and its applications. Outline the application of forging, rolling and additive manufacturing

1 Understanding and relate its applications. Machine Tools And Operations Lathe - constructional features and working, various lathe operations such as facing, centering, drilling, plain turning, boring, reaming, chamfering taper turning and thread cutting. - Shaper - constructional features and working of the crank and slotted lever type quick return motion mechanism - constructional features of a drilling machine. Hole making operations - drilling - reaming - boring. Milling - Column and knee type milling machine - various operations performed on milling machine. Constructional features of CNC machines, illustrate flexible manufacturing system and its applications. Outline the application of forging, rolling and additive manufacturing is an official topic in Module 3 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding and relate its applications. Machine Tools And Operations Lathe - constructional features and working, various lathe operations such as facing, centering, drilling, plain turning, boring, reaming, chamfering taper turning and thread cutting. - Shaper - constructional features and working of the crank and slotted lever type quick return motion mechanism - constructional features of a drilling machine. Hole making operations - drilling - reaming - boring. Milling - Column and knee type milling machine - various operations performed on milling machine. Constructional features of CNC machines, illustrate flexible

manufacturing system and its applications. Outline the application of forging, rolling and additive manufacturing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding and relate its applications. machine tools and operations lathe - constructional features and working, various lathe operations such as facing, centering, drilling, plain turning, boring, reaming, chamfering taper turning and thread cutting. - shaper - constructional features and working of the crank and slotted lever type quick return motion mechanism - constructional features of a drilling machine. hole making operations - drilling - reaming - boring. milling - column and knee type milling machine - various operations performed on milling machine. constructional features of cnc machines, illustrate flexible manufacturing system and its applications. outline the application of forging, rolling and additive manufacturing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding and relate its applications. Machine Tools And Operations Lathe - constructional features and working, various lathe operations such as facing, centering, drilling, plain turning, boring, reaming, chamfering taper turning and thread cutting. - Shaper - constructional features and working of the crank and slotted lever type quick return motion mechanism - constructional features of a drilling machine. Hole making operations - drilling - reaming - boring. Milling - Column and knee type milling machine - various operations performed on milling machine. Constructional features of CNC machines, illustrate flexible manufacturing system and its applications. Outline the application of forging, rolling and additive manufacturing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 1 Understanding and relate its applications. Machine Tools And Operations Lathe - constructional features and working, various lathe operations such as facing, centering, drilling, plain turning, boring, reaming, chamfering taper turning and thread cutting. - Shaper - constructional features and working of the crank and slotted lever type quick return motion mechanism - constructional features of a drilling machine. Hole making operations - drilling - reaming - boring. Milling - Column and knee type milling machine - various operations performed on milling machine. Constructional features of CNC machines, illustrate flexible manufacturing system and its applications. Outline the application of forging, rolling and additive manufacturing □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
milling, drilling machine and the basics of flexible manufacturing system.

            Identify the components of a lathe and define
  M3.01     their functions, working and lathe                  2
Understanding
            specifications.
            Define various lathe operations such as facing,
            centering, drilling, plain turning, boring,
                                                                1
Understanding
  M3.02     reaming, chamfering taper turning and thread
            cutting.
            Identify the components of a of shaper and
            demonstrate the crank and slotted lever type        1
Understanding
  M3.03
            quick return motion mechanism.

  M3.04     Identify the components of a drilling machine.      1
Understanding

  M3.05     Define drilling, reaming and boring.                1
Understanding

            Identify the components of a Column and
  M3.06                                                         2
Understanding
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Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

:

This module is presented from the official syllabus scope for Manufacturing Technology for. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: :
- Official duration: As specified
- Official topic entries captured: 9
- The full source excerpt is preserved below for coverage checking.

general classification of metal forming 1 Understanding operations.

general classification of metal forming 1 Understanding operations. is an official topic in Module 4 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify general classification of metal forming 1 Understanding operations. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, general classification of metal forming 1 understanding operations. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What general classification of metal forming 1 Understanding operations. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-□□ general classification of metal forming 1 Understanding operations. □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

Define forging. 1 Understanding Define forging, smith forging and machine

Define forging. 1 Understanding Define forging, smith forging and machine is an official topic in Module 4 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Define forging. 1 Understanding Define forging, smith forging and machine using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, define forging. 1 understanding define forging, smith forging and machine should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Define forging. 1 Understanding Define forging, smith forging and machine means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-[illegible] Define forging. 1 Understanding Define forging, smith forging and machine [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

1 Understanding forging Demonstrate the machine forging of crank

1 Understanding forging Demonstrate the machine forging of crank is an official topic in Module 4 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding forging Demonstrate the machine forging of crank using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding forging demonstrate the machine forging of crank should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding forging Demonstrate the machine forging of crank means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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but different term.

Malayalam support: Module 4-[illegible] 1 Understanding forging Demonstrate the machine forging of crank [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

1 Understanding shaft and connecting rod. Illustrate rolling and rolling mill

1 Understanding shaft and connecting rod. Illustrate rolling and rolling mill is an official topic in Module 4 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding shaft and connecting rod. Illustrate rolling and rolling mill using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding shaft and connecting rod. illustrate rolling and rolling mill should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding shaft and connecting rod. Illustrate rolling and rolling mill means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-□□ 1 Understanding shaft and connecting rod. Illustrate rolling and rolling mill □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

1 Understanding configurations. Define direct extrusion, indirect extrusion, hot extrusion, cold extrusion, continuous

1 Understanding configurations. Define direct extrusion, indirect extrusion, hot extrusion, cold extrusion, continuous is an official topic in Module 4 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding configurations. Define direct extrusion, indirect extrusion, hot extrusion, cold extrusion, continuous using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding configurations. define direct extrusion, indirect extrusion, hot extrusion, cold extrusion, continuous should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding configurations. Define direct extrusion, indirect extrusion, hot extrusion, cold extrusion, continuous means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a

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Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] 1 Understanding configurations. Define direct extrusion, indirect extrusion, hot extrusion, cold extrusion, continuous [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

1 Understanding extrusion, discrete extrusion and hydrostatic extrusion

1 Understanding extrusion, discrete extrusion and hydrostatic extrusion is an official topic in Module 4 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding extrusion, discrete extrusion and hydrostatic extrusion using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding extrusion, discrete extrusion and hydrostatic extrusion should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding extrusion, discrete extrusion and hydrostatic extrusion means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-[illegible] 1 Understanding extrusion, discrete extrusion and hydrostatic extrusion [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Define wire drawing and tube drawing.

Define wire drawing and tube drawing. is an official topic in Module 4 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Define wire drawing and tube drawing. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, define wire drawing and tube drawing. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Define wire drawing and tube drawing. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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Malayalam support: Module 4-[illegible] Define wire drawing and tube drawing. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Define hydroforming and shot peening. 1 Understanding Summarize the fundamentals, applications,

Define hydroforming and shot peening. 1 Understanding Summarize the fundamentals, applications, is an official topic in Module 4 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Define hydroforming and shot peening. 1 Understanding Summarize the fundamentals, applications, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, define hydroforming and shot peening. 1 understanding summarize the fundamentals, applications, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Define hydroforming and shot peening. 1 Understanding Summarize the fundamentals, applications, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Malayalam support: Module 4-□□ Define hydroforming and shot peening. 1 Understanding Summarize the fundamentals, applications, □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

classifications of additive manufacturing and 1 Understanding define rapid prototyping. Contents Metal Forming And Powder Metallurgy Hot working, cold working and classification of metal forming operations. Forging- smith forging and machine forging. Machine forging of crank shaft and connecting rod. Rolling and rolling mill configurations. direct extrusion, indirect extrusion, hot extrusion, cold extrusion, continuous extrusion, discrete extrusion, hydrostatic extrusion. Wire drawing, tube drawing, hydroforming, shot peening, Fundamentals, classifications and applications of additive manufacturing and rapid Prototyping. Text / Reference: T/R Book Title/Author Elements of Workshop Technology Vol-I Manufacturing Process edition-By T1 Hajra Choudry. Elements of Workshop Technology Vol-II Manufacturing Process edition-By T2 Hajra Choudry R1 Work shop technology By R.S KHURMI & J.K GUPTA of S.CHAND &Co. Ltd Manufacturing Engineering and Technology by Seropekalpakjian, Steven R R2 Schmid Pearson Education-Delhi R3 Manufacturing Technology-1By P.C Sharma of S.CHAND Publications. R4 Engineering Materials by Er. R.K.RAJPUT of S.CHAND Publications Online Resources: Sl.No Website Link 1 www.nptel.ac.in/courses/112105126/36 2 www.youtube.com/watch?v=T5gjkYvMg8A 3 http://www.youtube.com/watch?v=TeBX6cKKHWY

classifications of additive manufacturing and 1 Understanding define rapid prototyping. Contents Metal Forming And Powder Metallurgy Hot working, cold working and classification of metal forming operations. Forging- smith forging and machine forging. Machine forging of crank shaft and connecting rod. Rolling and rolling mill configurations. direct extrusion, indirect extrusion, hot extrusion, cold extrusion, continuous extrusion, discrete extrusion, hydrostatic extrusion. Wire drawing, tube drawing, hydroforming, shot peening, Fundamentals, classifications and applications of additive manufacturing and rapid Prototyping. Text / Reference: T/R Book Title/Author Elements of Workshop Technology Vol-I Manufacturing Process edition-By T1 Hajra Choudry. Elements of Workshop Technology Vol-II Manufacturing Process edition-By T2 Hajra Choudry R1 Work shop technology By R.S KHURMI & J.K GUPTA of S.CHAND &Co. Ltd Manufacturing Engineering and Technology by Seropekalpakjian, Steven R R2 Schmid Pearson Education-Delhi R3 Manufacturing Technology-1By P.C Sharma of S.CHAND Publications. R4 Engineering Materials by Er. R.K.RAJPUT of S.CHAND Publications Online Resources: Sl.No Website Link 1 www.nptel.ac.in/courses/112105126/36 2 www.youtube.com/watch?v=T5gjkYvMg8A 3 http://www.youtube.com/watch?v=TeBX6cKKHWY is an official topic in Module 4 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

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Study points

- Define or identify classifications of additive manufacturing and 1 Understanding define rapid prototyping. Contents Metal Forming And Powder Metallurgy Hot working, cold working and classification of metal forming operations. Forging-smith forging and machine forging. Machine forging of crank shaft and connecting rod. Rolling and rolling mill configurations. direct extrusion, indirect extrusion, hot extrusion, cold extrusion, continuous extrusion, discrete extrusion, hydrostatic extrusion. Wire drawing, tube drawing, hydroforming, shot peening, Fundamentals, classifications and applications of additive manufacturing and rapid Prototyping. Text / Reference: T/R Book Title/Author Elements of Workshop Technology Vol-I Manufacturing Process edition-By T1 Hajra Choudry. Elements of Workshop Technology Vol-II Manufacturing Process edition-By T2 Hajra Choudry R1 Work shop technology By R.S KHURMI & J.K GUPTA of S.CHAND &Co. Ltd Manufacturing Engineering and Technology by Seropekalpakjian, Steven R R2 Schmid Pearson Education-Delhi R3 Manufacturing Technology-1By P.C Sharma of S.CHAND Publications. R4 Engineering Materials by Er. R.K.RAJPUT of S.CHAND Publications Online Resources: Sl.No Website Link 1 www.nptel.ac.in/courses/112105126/36 2 www.youtube.com/watch?v=T5gjkYvMg8A 3 http://www.youtube.com/watch?v=TeBX6cKKHWY using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
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Engineering or professional relevance

In Polytechnic practice, classifications of additive manufacturing and 1 understanding define rapid prototyping. contents metal forming and powder metallurgy hot working, cold working and classification of metal forming operations. forging- smith forging and machine forging. machine forging of crank shaft and connecting rod. rolling and rolling mill configurations. direct extrusion, indirect extrusion, hot extrusion, cold extrusion, continuous extrusion, discrete extrusion, hydrostatic extrusion. wire drawing, tube drawing, hydroforming, shot peening, fundamentals, classifications and applications of additive manufacturing and rapid prototyping. text / reference: t/r book title/author elements of workshop technology vol-i manufacturing process edition-by t1 hajra choudry. elements of workshop technology vol-ii manufacturing process edition-by t2 hajra choudry r1 work shop technology by r.s khurmi & j.k gupta of s.chand &co. ltd manufacturing engineering and technology by seropekalpakjian, steven r r2 schmid pearson education-delhi r3 manufacturing technology-1by p.c sharma of s.chand publications. r4 engineering materials by er. r.k.rajput of s.chand publications online resources: sl.no website link 1 www.nptel.ac.in/courses/112105126/36 2 www.youtube.com/watch?v=t5gjkyvmg8a 3 http://www.youtube.com/watch?v=tebx6ckkhwy should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What classifications of additive manufacturing and 1 Understanding define rapid prototyping. Contents Metal Forming And Powder Metallurgy Hot working, cold working and classification of metal forming operations. Forging- smith forging and machine forging. Machine forging of crank shaft and connecting rod. Rolling and rolling mill configurations. direct extrusion, indirect extrusion, hot extrusion, cold extrusion, continuous extrusion, discrete extrusion, hydrostatic extrusion. Wire drawing, tube drawing, hydroforming, shot peening, Fundamentals, classifications and applications of additive manufacturing and rapid Prototyping. Text / Reference: T/R Book Title/Author Elements of Workshop Technology Vol-I Manufacturing Process edition-By T1 Hajra Choudry. Elements of Workshop Technology Vol-II Manufacturing Process edition-By T2 Hajra Choudry R1 Work shop technology By R.S KHURMI & J.K GUPTA of S.CHAND &Co. Ltd Manufacturing Engineering and Technology by Seropekalpakjian, Steven R R2 Schmid Pearson Education-Delhi R3 Manufacturing Technology-1By P.C Sharma of S.CHAND Publications. R4 Engineering Materials by Er. R.K.RAJPUT of S.CHAND Publications Online Resources: Sl.No Website Link 1 www.nptel.ac.in/courses/112105126/36 2 www.youtube.com/watch?v=T5gjkYvMg8A 3 http://www.youtube.com/watch?v=TeBX6cKKHWY means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Malayalam support: Module 4-□□ classifications of additive manufacturing and 1 Understanding define rapid prototyping. Contents Metal Forming And Powder Metallurgy Hot working, cold working and classification of metal forming operations. Forging- smith forging and machine forging. Machine forging of crank shaft and connecting rod. Rolling and rolling mill configurations. direct extrusion, indirect extrusion, hot extrusion, cold extrusion, continuous extrusion, discrete extrusion, hydrostatic extrusion. Wire drawing, tube drawing, hydroforming, shot peening, Fundamentals, classifications and applications of additive manufacturing and rapid Prototyping. Text / Reference: T/R Book Title/Author Elements of Workshop Technology Vol-I Manufacturing Process edition-By T1 Hajra Choudry.

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Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
:
             techniques in automobile industry.
             Summarize hot working, cold working and the
  M4.01      general classification of metal forming            1
Understanding
             operations.

  M4.02      Define forging.                                    1
Understanding

             Define forging, smith forging and machine
  M4.03                                                         1
Understanding
             forging
             Demonstrate the machine forging of crank
  M4.04                                                         1
Understanding
             shaft and connecting rod.

             Illustrate rolling and rolling mill
  M4.05                                                         1
Understanding
             configurations.
             Define direct extrusion, indirect extrusion, hot
             extrusion, cold extrusion, continuous
  M4.06                                                         1
Understanding
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
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Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Recall casting and pattern. 1 Remembering Outline the requirements of a pattern material. (3 marks)

Answer: Begin with the standard definition or identity of *Recall casting and pattern. 1 Remembering Outline the requirements of a pattern material*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 1 Understanding and list different types of pattern materials. Classify patterns like single, split, loose piece. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding and list different types of pattern materials. Classify patterns like single, split, loose piece*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 1 Understanding and sweep pattern. Define pattern allowance and define pattern. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding and sweep pattern. Define pattern allowance and define pattern*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain allowances such as shrinkage, machining, 1 Understanding draft, distortion, shake allowances. Define mould, cope, drag, runner, riser, core. (3 marks)

Answer: Begin with the standard definition or identity of *allowances such as shrinkage, machining, 1 Understanding draft, distortion, shake allowances. Define mould, cope, drag, runner, riser, core*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain 1 Understanding application of sheet metal in automobile. List metals used for sheet metal work and. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding application of sheet metal in automobile. List metals used for sheet metal work and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 1 Understanding standard gauge numbers used for sheet metals. Define various sheet metal operations like - shearing, cutting off, parting, blanking,. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding standard gauge numbers used for sheet metals. Define various sheet metal operations like - shearing, cutting off, parting, blanking,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 1 Understanding punching, notching, slitting, lancing, bending, drawing and squeezing.. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding punching, notching, slitting, lancing, bending, drawing and squeezing.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Define welding and weldability. 1 Remembering Explain the features of welding and general. (3 marks)

Answer: Begin with the standard definition or identity of *Define welding and weldability. 1 Remembering Explain the features of welding and general*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain their functions, working and lathe 2 Understanding specifications. Define various lathe operations such as facing, centering, drilling, plain turning, boring,. (3 marks)

Answer: Begin with the standard definition or identity of *their functions, working and lathe 2 Understanding specifications. Define various lathe operations such as facing,* *centering, drilling, plain turning, boring,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain reaming, chamfering taper turning and thread cutting. Identify the components of a of shaper and demonstrate the crank and slotted lever type 1 Understanding M3.03 quick return motion mechanism.. (3 marks)

Answer: Begin with the standard definition or identity of *reaming, chamfering taper turning and thread cutting. Identify the components of a of shaper and demonstrate the crank and slotted lever type 1 Understanding M3.03 quick return motion mechanism.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Identify the components of a drilling machine.. (3 marks)

Answer: Begin with the standard definition or identity of *Identify the components of a drilling machine.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Define drilling, reaming and boring. 1 Understanding Identify the components of a Column and. (3 marks)

Answer: Begin with the standard definition or identity of *Define drilling, reaming and boring. 1 Understanding Identify the components of a Column and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Module 4

Define or explain general classification of metal forming 1 Understanding operations.. (3 marks)

Answer: Begin with the standard definition or identity of *general classification of metal forming 1 Understanding operations.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Define forging. 1 Understanding Define forging, smith forging and machine. (3 marks)

Answer: Begin with the standard definition or identity of *Define forging. 1 Understanding Define forging, smith forging and machine*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 1 Understanding forging Demonstrate the machine forging of crank. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding forging Demonstrate the machine forging of crank*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 1 Understanding shaft and connecting rod. Illustrate rolling and rolling mill. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding shaft and connecting rod. Illustrate rolling and rolling mill*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Recall casting and pattern. 1 Remembering Outline the requirements of a pattern material**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **1 Understanding application of sheet metal in automobile. List metals used for sheet metal work and**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **their functions, working and lathe 2 Understanding specifications. Define various lathe operations such as facing, centering, drilling, plain turning, boring,**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **general classification of metal forming 1 Understanding operations.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link www.nptel.ac.in/courses/112105126/36 www.youtube.com/watch?v=T5gjkYvMg8A http://www.youtube.com/watch?v=TeBX6cKKHWY

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 3052. Verify institutional updates against the current official curriculum.